SegJointVP: Motion-Free Segmentation Conditioning for Video Prediction

Hanbom Woo [*,1], Hyuk Heo[*,2], Il Yong Chun[†,1,2,3]

[1]Department of Electrical and Computer Engineering, Sungkyunkwan University (SKKU)
[2]Department of Display Convergence Engineering, SungKyunKwan University (SKKU)
[3]Department of AI, ADE, and SCE, SKKU
[*]These authors contributed equally.
[†]Corresponding author

mongyong2@g.skku.edu, sgcwhb@g.skku.edu,iychun@skku.edu

Abstract

Predicting future video frames from limited observations is a challenging task due to the high uncertainty of motion and appearance dynamics. We propose **SegJointVP**, a diffusion-based video prediction framework that leverages confidence-aware instance segmentation to guide future frame generation without relying on explicit motion prediction. SegJointVP extracts instance-level embeddings from segmentation maps of past video and refines them through the proposed **SegJoint** module, which incorporates entropy-based confidence scores to emphasize reliable regions while suppressing ambiguous areas. The confidence-modulated instance features are injected as conditioning signals into the EDM [1]-based denoising process, facilitating accurate and stable frame synthesis. Experiments on the Cityscapes [2] dataset demonstrate that SegJointVP consistently outperforms baseline methods, producing high-quality video predictions with better preservation of instance structures.

1. Introduction

Video prediction has become a critical component in many real-world applications, including autonomous driving [3] and robotic navigation [4], where anticipating future visual observations is essential for safe and robust decision-making. Despite its importance, predicting future frames remains inherently challenging due to the limited information available in a short sequence of input frames and the high uncertainty of future dynamics.

To address this difficulty, recent studies have explored the use of additional cues beyond RGB image, such as optical flow, motion trajectories, or depth, to guide future frame synthesis. While explicitly predicting future motion can provide strong inductive biases, it also introduces a critical limitation: errors in motion estimation tend to accumulate over time, leading to increasingly inaccurate predictions, especially in regions with complex or ambiguous dynamics.

Motivated by this observation, we propose **SegJointVP**, a video prediction model that conditions future frames on instance-level segmentation embeddings without explicitly predicting motion.

SegJointVP incorporates the proposed novel **SegJoint** module, which extracts instance embeddings from segmentation maps and selectively applies instance-based conditioning to reliable regions while suppressing ambiguous areas. By leveraging entropy-based confidence scores, avoiding propagating uncertain signals and instead focuses conditioning on regions where instance cues are trustworthy.

Our framework mitigates error accumulation and enables the model to focus on more trustworthy regions, resulting in more stable and accurate future frame generation.

2. Backgrounds

This section provides a brief overview of diffusion models. Diffusion models [5] are characterized by a pair of forward and reverse processes. In the forward process, isotropic Gaussian noise is progressively injected into a clean sample x_0, producing a noisy sample $x_\sigma = x_0 + \sigma\epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. The noise standard deviation σ is determined by a predefined monotonically increasing schedule, ranging from $\sigma_{\min}$ to $\sigma_{\max}$, which denote the minimum and maximum noise levels, respectively.

The reverse process starts from a pure noise sample $x_{\sigma_{\max}} \sim \mathcal{N}(0, \sigma_{\max}^2 I)$ and progressively denoises it using a learned denoiser $D_\theta(x_\sigma, \sigma)$ parameterized by θ, eventually reconstructing a clean sample x_0. Following the EDM framework [1], the denoiser is trained by minimizing the objective:

$$\mathbb{E}_{x_0,\epsilon,\sigma}, \lambda_\sigma \| D_\theta(x_\sigma, \sigma) - x_0 \|_2^2, \tag{1}$$

where λ_σ is a weighting function that depends on the noise level. To further stabilize training across different noise scales, EDM adopts a preconditioning scheme that rescales both the input and output of the denoising network. Under this formulation, the denoiser is expressed as [1]:

$$D_\theta(x_\sigma, \sigma) = c_{\text{out}}(\sigma)\, D'_\theta\big(c_{\text{in}}(\sigma)x_\sigma, c_{\text{noise}}(\sigma)\big) + c_{\text{skip}}(\sigma)x_\sigma, \tag{2}$$

where D'_θ denotes the underlying neural network, and $c_{\text{in}}(\sigma)$, $c_{\text{out}}(\sigma)$, $c_{\text{noise}}(\sigma)$, and $c_{\text{skip}}(\sigma)$ are predefined preconditioning coefficients that modulate the network behavior at different noise levels.

2.1. Method

This section describes the methodology of **SegJointVP**. Figure 1 illustrates the overall architecture of the proposed model.

Given M input RGB frames $\mathbf{X}^0_{1:M} \in \mathbb{R}^{M\times H\times W\times 3}$ and their corresponding segmentation maps $\mathbf{S}^0_{1:M} \in \mathbb{R}^{(K+1)\times M\times H\times W}$, the model predicts N future frames $\mathbf{X}_{M+1:M+N} \in \mathbb{R}^{N\times H\times W\times 3}$. We denote the video total sequence length by T, where $T = M + N$, and H, W are the height and width of the image.

From the instance segmentation maps, we select the K most prominent instances and the background, resulting in a total of $K+1$ instances. For each instance, an instance-level embedding is extracted, forming a set of embeddings $\mathbf{I}_{1:K+1} \in \mathbb{R}^{(K+1)\times D}$, where D is the channel size of the embedding. These embeddings are then fed into the proposed **SegJoint** module $\mathcal{J}$, which refines and projects them using entropy-based confidence scores. The confidence-modulated instance features are subsequently injected as conditioning signals into the EDM-based denoising process, supporting stable and reliable instance-level structure in future frames.

2.1.1 Instance correlation map

We employ a pretrained instance segmentation model, Uni-DVPS [6], to generate pseudo-labels. During training, we identify the top-K instances that occupy the largest areas in the first frame and track these instances across all subsequent frames. For each frame, we extract instance-level embeddings for the K selected instances as well as the background, resulting in a total of $K + 1$ embeddings. These embeddings $\mathbf{I}_{1:K+1}$ are later used to compute confidence scores, while the instance segmentation maps serve as pseudo supervision.

Based on the instance segmentation maps, we extract instance-level embeddings and compute their similarity to the intermediate denoising features along the denoising path. Let $\mathbf{d}_t \in \mathbb{R}^{T\times H'\times W'\times C}$ denote the intermediate feature map at frame index $t \in \{1, \ldots, T\}$, where $H' \times W'$ corresponds to the spatial resolution of the features. This similarity computation yields instance correlation maps:

$$\mathbf{c}_t = \langle \mathcal{R}(\mathbf{I}_{1:K+1}),\ \mathbf{d}_t \rangle, t \in \{1, \ldots, T\} \tag{3}$$

where $\langle \cdot, \cdot \rangle$ denotes an inner-product–based similarity measure, and $\mathcal{R}$ represent the refiner module, implemented as MLP (Multi-layer perceptron) layers and ReLU activation function for feature alignment and match same feature dimension with instance embeddings.

2.1.2 Proposed SegJoint module

The instance correlation map $\mathbf{c}_{1:T}$ is interpreted as a probability-like map and serves two purposes: it provides weighting factors that determine the extent to which instance embeddings are incorporated, and it is further used to compute entropy-based confidence scores.

As illustrated in Figure 1, the instance correlation map $\mathbf{c}_t$ is first normalized using a softmax operation to obtain a weighting map:

$$\mathbf{w}_t = \text{softmax}\,(\mathbf{c}_t), \tag{4}$$

which modulates the contribution of instance embeddings in a probabilistic manner.

Using this weighting map, an instance feature map is constructed as a weighted combination of instance embeddings. The instance feature map $\mathbf{f}_t \in \mathbb{R}^{T\times H'\times W'\times D}$ is computed as:

$$\mathbf{f}_t = \sum_{k=1}^{K+1} \mathbf{w}_t^{(k)} \odot \mathbf{I}_{1:K+1}, \tag{5}$$

where $\odot$ denotes broadcasted element-wise multiplication over spatial dimensions. The resulting instance feature map $\mathbf{f}_t$ is then projected through a projector $\mathcal{P}$ to align it with the denoising video features and added to the denoising features.

To obtain the confidence map, the normalized correlation map is first upsampled and used to compute entropy-based scores. Specifically, let $\tilde{\mathbf{c}}_t$ denote the upsampled correlation map. The confidence map $\mathbf{q}_t$ is computed as:

$$\mathbf{q}_t = -\sum_{k=1}^{K+1} \tilde{\mathbf{c}}_t^{(k)} \log \tilde{\mathbf{c}}_t^{(k)}, \tag{6}$$

where higher entropy indicates lower confidence in instance association.

The confidence map is then passed through MLP layers to obtain the weighting factor $\alpha \in \mathbb{R}^{T\times H'\times W'\times 1}$:

$$\alpha = MLP(\mathbf{q}_t), \tag{7}$$

which is subsequently used in the Lerp operation to modulate the contribution of instance features.

Confidence awareness is incorporated by transforming the confidence map through a linear layer and using it as a weighting factor in a linear interpolation (Lerp) operation:

$$\mathbf{d}'_t = (1 - \alpha) \odot \mathbf{d}_t + \alpha \odot \mathcal{P}(\mathbf{f}_t), \tag{8}$$

This formulation ensures that regions with high entropy (low confidence) have reduced influence on the fused features.

The confidence-aware feature fusion is performed on a patch-wise basis at the first decoder block of the denoising path, allowing reliable instance features to guide generation while suppressing ambiguous regions.

We detach the gradients of the instance feature maps and confidence maps before passing them into the SegJoint module, such that the module can focus solely on learning reliable instance-level correspondence priors without interfering with the main denoising optimization.

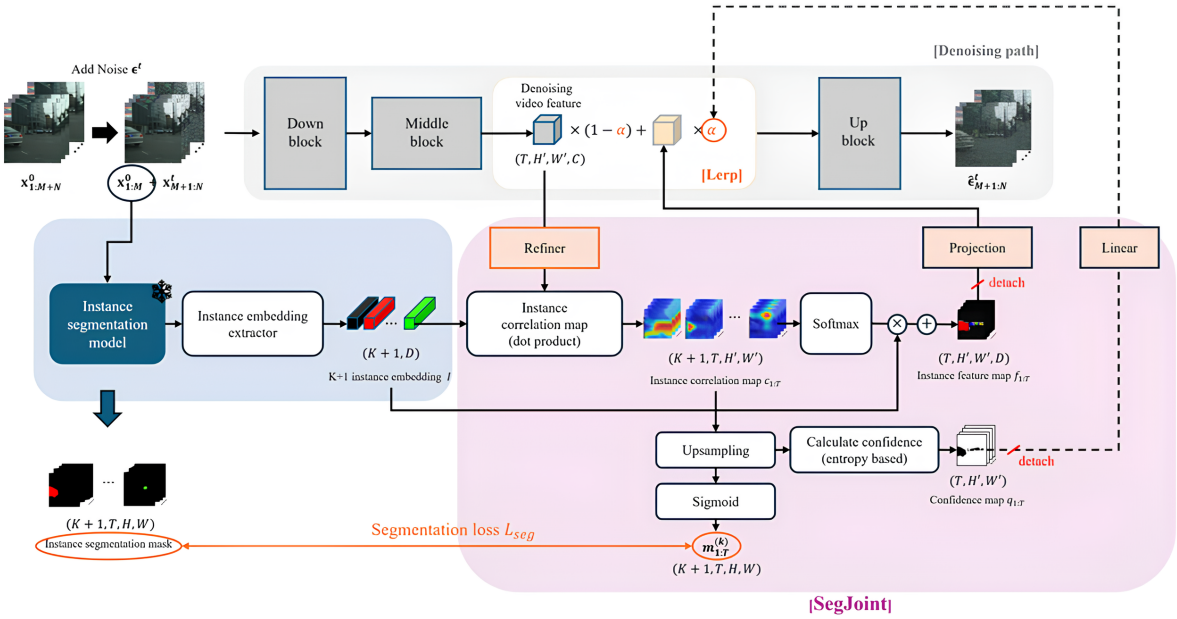


Figure 1. Architecture of proposed **SegJointVP**.

2.2. Results and Discussion

In this section, we describe about experimental setups and present the results with some discussion of our models. In addition, we compared state-of-the art (SOTA) video prediction model WALDO [7].

2.2.1 Dataset and evaluation metrics

We conduct experiments on the Cityscapes [2] dataset. During training, the model takes $M = 4$ input frames and predicts $N = 8$ future frame tuples. For inference, we evaluate on the Cityscapes test set by first predicting $N = 8$ future frames given $M = 4$ input frames. The predicted frames are then used in an autoregressive manner with an overlap of 4 frames, repeating the process three times to generate a total of 28 future frames. We set $K = 8$ for the number of maximum instances.

We used peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), learned perceptual image patch similarity (LPIPS) and Fréchet video distance (FVD) for the evaluation metrics, which is an comprehensive setup in video prediction.

2.2.2 Implementation details

The model operates on inputs with a spatial resolution of $H = 128$ and $W = 128$, and consists of 3 encoder blocks, 8 middle blocks, and 3 decoder blocks. The proposed SegJoint module is applied at an intermediate resolution of $H' = 32$ and $W' = 32$, where instance-level semantics are sufficiently abstracted while preserving spatial structure. The model is trained with an objective composed of a reconstruction loss and a segmentation-guided regularization:

$$\mathcal{L} = \mathcal{L}_{\text{mse}} + 0.01\mathcal{L}_{\text{seg}}. \tag{9}$$

Segmentation Loss. We first upsample the correlation map $c_{1:T}$ to the video resolution, and apply a sigmoid activation to obtain binary instance mask $m \in \mathbb{R}^{(K+1)\times H\times W}$ for $K+1$ instances. The segmentation loss is then computed between the predicted mask m and ground-truth masks m^{gt} using a linear combination of Dice loss and Focal loss, where Focal term is weighted by $\lambda_{focal} = 20$.

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{dice}}(m, m^{\text{gt}}) + \lambda_{\text{focal}}\mathcal{L}_{\text{focal}}(m, m^{\text{gt}}) \tag{10}$$

2.2.3 Discussion of Segjoint module

Method	PSNR ↑	SSIM ↑	LPIPS ↓	FVD ↓
w/o SegJoint	23.34	0.708	0.097	106.5
w/ SegJoint	**23.76**	**0.719**	**0.093**	**100.62**

Table 1. Quantitative comparison with and without SegJoint.

As shown in Table 1 and Figure 2, the proposed SegJoint module effectively learns segmentation-related representations and accurately identifies reliable instance regions. By injecting instance embeddings into high-confidence areas, the model generates clearer and more consistent instance structures in the predicted frames. Compared to the baseline without SegJoint, our approach achieves consistent improvements across all evaluation metrics, demonstrating its superior predictive performance. Moreover, the instance embeddings extracted from the initial frames are reused throughout the entire autoregressive prediction process, enabling stable feature

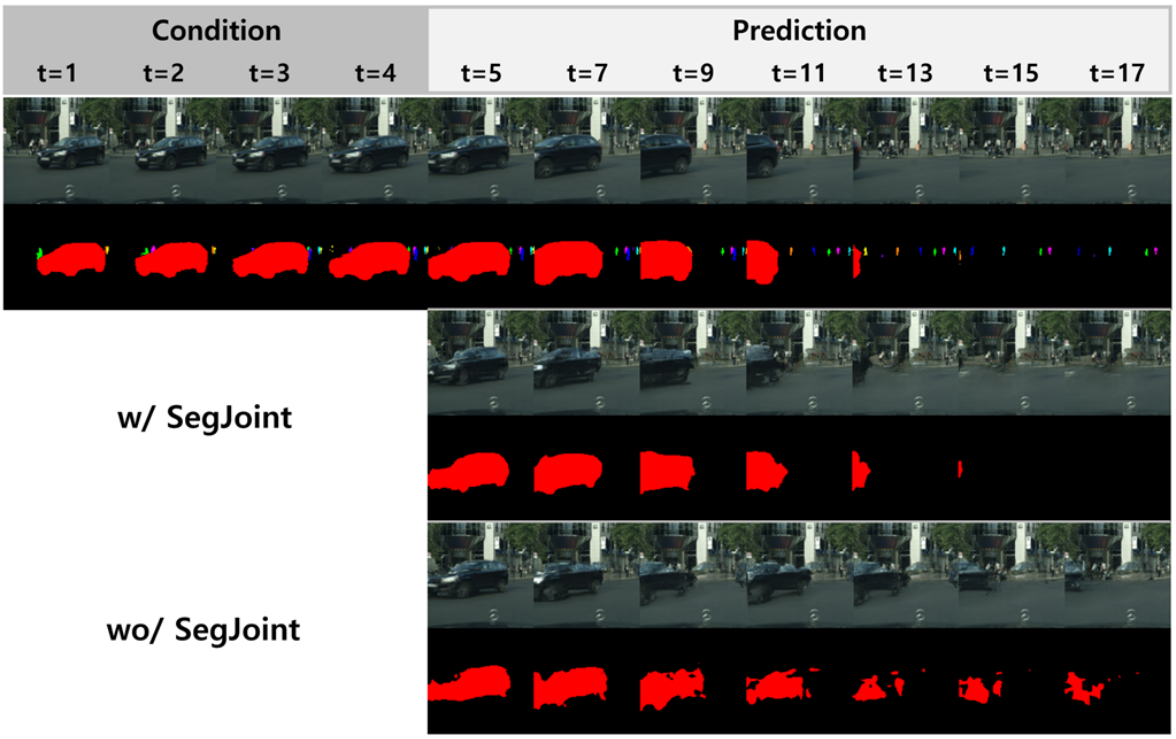


Figure 2. Qualitive comparison in Segjoint module.

sharing across prediction steps. This design mitigates feature drift during autoregressive generation and preserves temporal consistency, resulting in more stable and coherent long-term video predictions.

2.2.4 Comparison between video prediction method

Under the same experimental conditions, SegJointVP consistently outperforms WALDO across all reported metrics, including PSNR, SSIM, and LPIPS. As shown in Table 2, SegJointVP achieves higher reconstruction fidelity and perceptual quality.

Method	PSNR ↑	SSIM ↑	LPIPS ↓
WALDO	22.40	0.653	0.147
SegJointVP	**23.76**	**0.719**	**0.093**

Table 2. Comparison with WALDO under the same experimental settings. FVD results for WALDO are not reported in this setting.

3. Conclusion

We proposed SegJointVP, a motion-free diffusion-based video prediction framework that conditions future generation on instance-level segmentation embeddings. By introducing the SegJoint module with entropy-based confidence-aware conditioning, our approach selectively emphasizes reliable regions while suppressing ambiguous areas, effectively mitigating error accumulation in autoregressive prediction.

4. Acknowledgments

This work was supported in part by NRF grants 2022R1F1A-1074546 and RS-2023-00213455 funded by MSIT, the KEIT Technology Innovation program grant 20014967 funded by MOTIE, the BK21 FOUR Project, IITP grant 2019-0-00421 funded by MSIT, IBS grant R015-D1, KIAT grant P0022098 funded by MOTIE, SKKU-SMC and SKKU-KBSMC Future Convergence Research Program grants, and SKKU seed grants.

References

[1] T. Karras, M. Aittala, T. Aila, and S. Laine, "Elucidating the design space of diffusion-based generative models," *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[2] M. Cordts, M. Omran, S. Ramos, T. Rehfeld, R. B. Markus Enzweiler, U. Franke, S. Roth, and B. Schiele, "The cityscapes dataset for semantic urban scene understanding," *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[3] N. J. Merin Abraham, Nikita Suryawanshi and D. Hadsul, "Future predicting intelligent camera security system," 2021.

[4] A. Bhattacharyya, M. Fritz, and B. Schiele, "Long-term on-board prediction of people in traffic scenes under un-

certainty," *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[5] A. J. Ho and P. Abbeel, "Denoising diffusion probabilistic models," *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[6] K. Ji-Yeon, O. Hyun-Bin, K. Byung-Ki, D. Kim, Y. Kwon, and T.-H. Oh, "Uni-dvps: Unified model for depth-aware video panoptic segmentation," *International Conference on Intelligent Robots and Systems (IROS)*, 2024.

[7] G. L. Moing, J. Ponce, and C. Schmid, "Waldo: Future video synthesis using object layer decomposition and parametric flow prediction," *IEEE International Conference on Computer Vision (ICCV)*, 2023.